Matrices & Vectors

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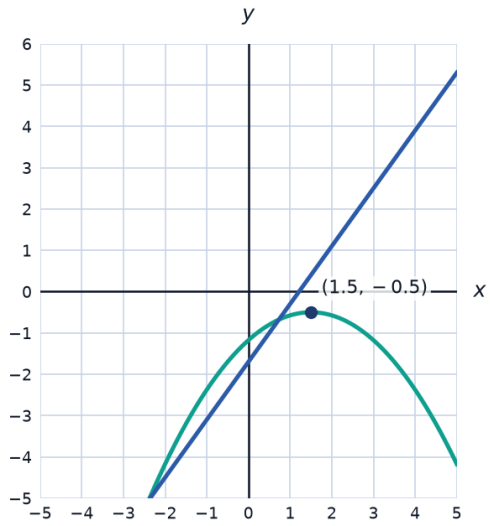
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and B be the matrix with rows $\langle 2, -1\rangle$ and $\langle 0, 5\rangle$. Find $A + B$ and $2A$.

2 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 1, 3\rangle$, $\langle -2, 4\rangle$) and B (rows $\langle 2, -1\rangle$, $\langle 0, 5\rangle$) from above, find AB.



3 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and let $\mathbf{v} = \langle 3, -1\rangle$. Find $A\mathbf{v}$.

4 **[Foundational]** • **Target: Determinant and inverse**

Find $\det A$ for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and state whether A is invertible.

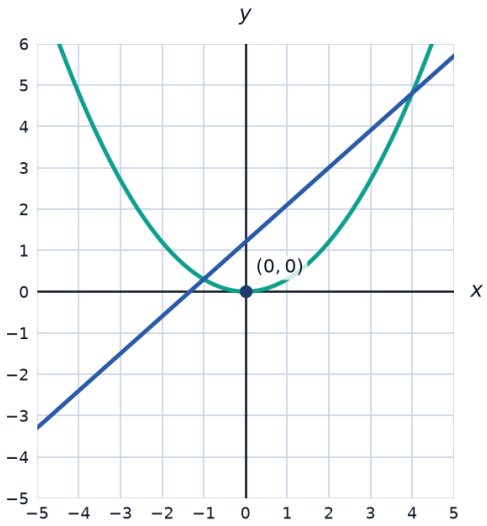
5 **[Foundational]** • **Target: Determinant and inverse**

Find A^{-1} for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$.

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6 [Foundational] • Target: Solving systems with the inverse matrix

Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).



7 [Foundational] • Target: Angle between vectors

Find the angle between $\mathbf{p} = \langle 2, 2 \rangle$ and $\mathbf{q} = \langle 3, 0 \rangle$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Angle between vectors

Find the angle between $\langle 7, 1 \rangle$ and $\langle 2, -3 \rangle$, rounded to the nearest tenth of a degree.

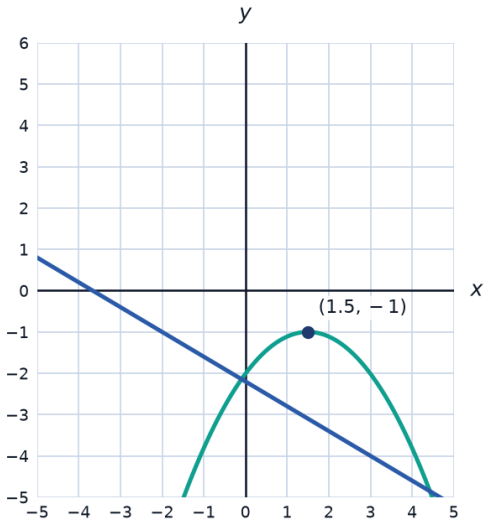
9 [Intermediate] • Target: Determinant and inverse

Given 3D vectors $\mathbf{u} = \langle 2, -3, 6 \rangle$ and $\mathbf{v} = \langle 1, 4, 2 \rangle$: (a) Find the unit vector in the direction of $\mathbf{u}$. (b) Calculate the angle θ between $\mathbf{u}$ and $\mathbf{v}$ to the nearest degree.

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10 **[Intermediate]** • **Target: Solving systems with the inverse matrix**

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.



11 **[Intermediate]** • **Target: Vector magnitude and dot product**

Find all real values of the constant k for which the matrix $M = (k + 2, 4; 3, k - 2)$ is singular (has no inverse).

12 **[Intermediate]** • **Target: Angle between vectors**

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

13 **[Intermediate]** • **Target: Addition and scalar multiplication**

Find the magnitude and direction angle θ (in degrees, $0^\circ \le \theta < 360^\circ$) of the vector $\mathbf{v} = \langle -5, 5\sqrt{3} \rangle$.

14 **[Intermediate]** • **Target: Matrix multiplication**

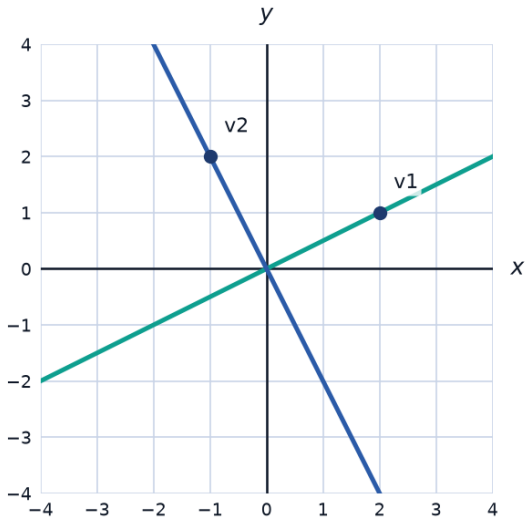
Find the angle θ between the vectors $\mathbf{a} = \langle 1, 2, 2 \rangle$ and $\mathbf{b} = \langle 2, -2, 1 \rangle$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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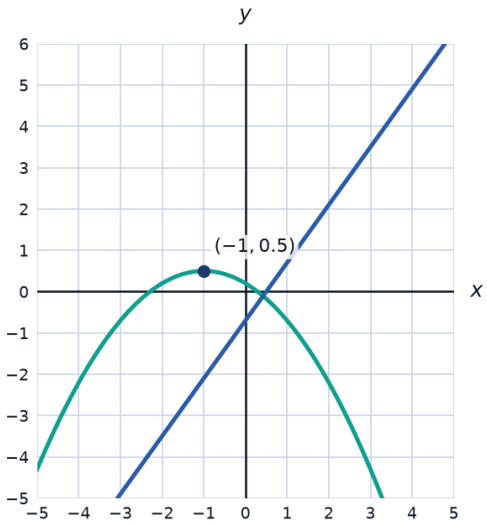
15 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 3.01 Functions & Graphs

Find the inverse of the matrix $A = (4, 7; 1, 2)$.



16 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 5.01 Introductory Calculus

Compute the matrix product AB where $A = (1, 3; -2, 4)$ and $B = (2, 0; 5, -1)$.



17 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 6.01 Probability

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

18 [Challenging] • Target: Angle between vectors • Cross-Lesson: 1.01 Complex Numbers

Vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 6, -8 \rangle$ are perpendicular. Find the value of k, and compute the projection of $\mathbf{c} = \langle 10, 5 \rangle$ onto $\mathbf{b}$.

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19 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 2.01 Sequences & Series

Let $A = (3, 1; 5, 2)$. Find A^{-1} and use it to solve the linear system: $3x + y = 7$ $5x + 2y = 12$

20 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 3.01 Functions & Graphs

Calculate the exact value of $\cos(75^\circ)$ using angle addition, and evaluate $\tan(2\theta)$ given that $\tan\theta = \frac{2}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Addition and scalar multiplication

Consider a linear transformation T represented by the matrix $A = (2, 1\ 1, 3)$. Let $P = (x; y)$ be a point in the plane. The transformation maps P to P'. If the area of the region transformed by T from a unit square in the xy-plane is A, and the determinant of A is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is 5.

a. 5

b. 2

c. 1

d. 10

22 [Foundational] • Target: Matrix multiplication

Consider the matrix $A = (2, 0; 0, 3)$. When the matrix A is applied to the vector $\mathbf{v} = (x; y)$, the resulting vector is $A\mathbf{v} = (6; 9)$. What is the original vector $\mathbf{v}$?

a. $(3; 3)$

b. $(1; 2)$

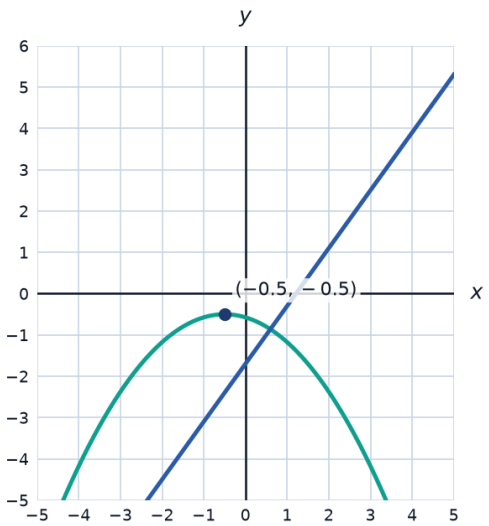
c. $(6; 3)$

d. $(12; 27)$

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23 [Foundational] • Target: Determinant and inverse

Find the angle θ between $\mathbf{u} = \langle 1, 7 \rangle$ and $\mathbf{v} = \langle -7, 4 \rangle$, to the nearest 0.1°.



a. 68.4°

b. 111.6°

c. 78.4°

d. 63.4°

24 [Foundational] • Target: Solving systems with the inverse matrix

Solve using the inverse-matrix method: $\{-4x + 5y = 17; 2x + 2y = -4$.

a. x=-3,\ y=1

b. x=1,\ y=-3

c. x=-2,\ y=1

d. x=-3,\ y=-1

25 [Foundational] • Target: Vector magnitude and dot product

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the coordinate plane such that $|\mathbf{u}| = 5$, $|\mathbf{v}| = 10$, and the angle θ between them is obtuse ($90^\circ < \theta < 180^\circ$). If the minimum value of $|t\mathbf{u} + \mathbf{v}|$ over all real numbers t is equal to 6, what is the value of the dot product $\mathbf{u} \cdot \mathbf{v}$?

a. -50

b. -40

c. -30

d. 40

26 [Foundational] • Target: Angle between vectors

A matrix $M = (k, 2; 3, 1)$ transforms a point $P(1, 2)$ to $P'(5, 5)$. Find the value of k.

a. 1

b. 2

c. 3

d. 4

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27 [Foundational] • Target: Addition and scalar multiplication

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

a. [3, 4]

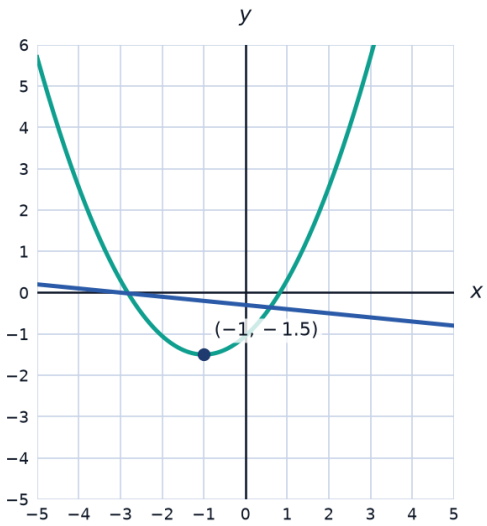
b. [5, 8]

c. [-3, -4]

d. [4, 3]

28 [Foundational] • Target: Matrix multiplication

In the standard (x, y)-coordinate plane, vectors $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ have magnitudes $||\mathbf{u}|| = 13$ and $||\mathbf{v}|| = 5$. If their dot product is $\mathbf{u} \cdot \mathbf{v} = 25$, and the matrix $M = [a, b; c, d]$ has $\det(M) > 0$, what is the value of $\det(M)$?



a. 25

b. 40

c. 60

d. 65

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Determinant and inverse

For all real values of θ, the matrix product $[\cos\theta, \sin\theta][6, 2; 2, 3][\cos\theta; \sin\theta]$ evaluates to a 1×1 matrix $[k]$. What is the maximum possible value of k?

a. 6

b. 7

c. 9

d. 14

30 [Intermediate] • Target: Solving systems with the inverse matrix

Matrices A and B are defined as $A = [2, k; -1, 3]$ and $B = [1, 4; 2, k]$, where k is a real number. If the determinant of the matrix sum $A + B$ is equal to 17, what is the value of k? (Note: The determinant of a 2×2 matrix $[a, b; c, d]$ is defined as $ad - bc$.)

a. 1

b. 2

c. 6

d. $\frac{19}{2}$

31 [Intermediate] • Target: Vector magnitude and dot product

For a 2×2 matrix $[a, b; c, d]$, the determinant is defined as $ad - bc$. Given the matrices $A = [x, 2; -1, 3]$ and $B = [1, -4; 3, x]$, the determinant of the matrix $A + B$ is equal to 19. If $x > 0$, what is the value of x?

a. 2

b. $\frac{5}{4}$

c. 4

d. 6

32 [Intermediate] • Target: Angle between vectors

In the standard coordinate plane, vector $\mathbf{u} = \langle 2, -1 \rangle$ and vector $\mathbf{v} = \langle k, 5 \rangle$, where k is a real constant. If the magnitude of the vector $\mathbf{u} + \mathbf{v}$ is equal to 5, which of the following is a possible value of k?

a. -5

b. -1

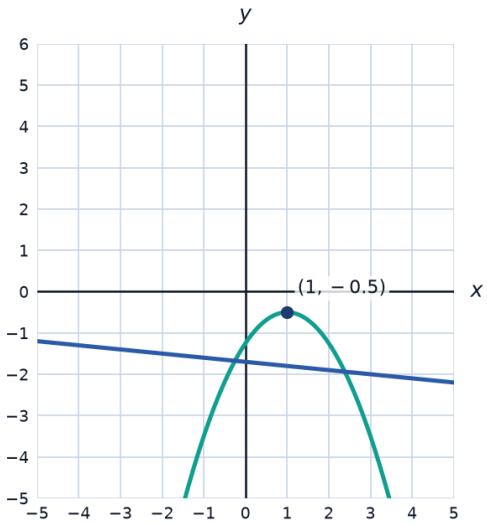
c. 3

d. 7

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33 **[Intermediate]** • **Target: Addition and scalar multiplication**

What is the value of the determinant of the matrix $[7, -3; 2, 4]$?



a. 34

b. 22

c. -29

d. -34

34 **[Intermediate]** • **Target: Matrix multiplication**

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$

b. $\langle 1, 1 \rangle$

c. $\langle 4, 1 \rangle$

d. $\langle 8, -13 \rangle$

35 **[Intermediate]** • **Target: Determinant and inverse**

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

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36 [Intermediate] • Target: Solving systems with the inverse matrix

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. 5

b. 7

c. 12

d. 25

37 [Intermediate] • Target: Vector magnitude and dot product

If matrix $A = [3, -2; 0, 4]$ and matrix $B = [-1, 5; 2, -3]$, what is the matrix $2A + B$?

a. $[5, 1; 2, 5]$

b. $[2, 3; 2, 1]$

c. $[5, 3; 2, 1]$

d. $[7, -9; -2, 11]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle between vectors • Cross-Lesson: 1.01 Complex Numbers

Find the angle between $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$ (to one decimal place).

a. 0.0°

b. 20.0°

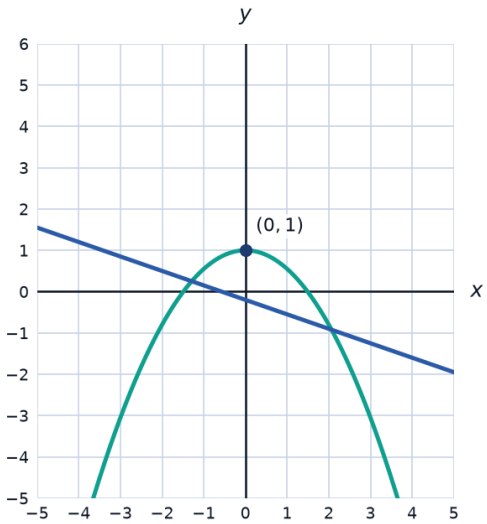
c. 180.0°

d. 10.0°

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39 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 2.01 Sequences & Series

A matrix has 1 rows and 3 columns. What are its dimensions?



a. 1×3

b. 4

c. 3

d. 3×1

40 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 3.01 Functions & Graphs

A vector $\mathbf{u} = \langle 3, k \rangle$ has magnitude 5. Find the positive value of k, given $k = 4$ is one solution.

a. 4

b. 2

c. 3

d. 5

41 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 5.01 Introductory Calculus

Find the determinant of $(-1, 2;\ -6, 4)$.

a. 8

b. -16

c. 9

d. 7

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42 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 6.01 Probability

A factory produces two types of components, A and B. Components of type A have a defect rate of 3%, and components of type B have a defect rate of 5%. A batch of 100 components is produced, consisting of N components of type A and $100 - N$ components of type B. Let X be the random variable representing the total number of defective components in the batch. If the expected number of defective components is $E[X] = 4.2$, find the probability that a randomly selected component from this batch is defective, given that it is of type A.

a. 0.030

b. 0.042

c. 0.050

d. 0.036

43 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 1.01 Complex Numbers

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

a. $\frac{3}{5}$

b. $\frac{2}{5}$

c. $\frac{1}{2}$

d. $\frac{3}{4}$

44 [Challenging] • Target: Angle between vectors • Cross-Lesson: 2.01 Sequences & Series

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

a. 1/6

b. 1/3

c. 1/2

d. 2/3

45 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 3.01 Functions & Graphs

A bag contains 5 red marbles and 3 blue marbles. If a marble is drawn at random, what is the probability that it is blue?

a. $\frac{3}{8}$

b. $\frac{5}{8}$

c. $\frac{3}{5}$

d. $\frac{5}{3}$

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